Stationary points

1

a Minimum turning point

b Maximum turning point

c Horizontal point of inflection

2

a

i $\frac{dy}{dx} = -12x + 84$

ii $(7, 265)$

iii $(7, 265)$ is a maximum turning point

b

i $\frac{dy}{dx} = 3x^2 - 42x + 144$

ii $(6, 305), (8, 301)$

iii $(6, 305)$ is a local maximum, and $(8, 301)$ is a local minimum

c

i $f'(x) = 3x^2 + 24x + 45$

ii $(-5, 4), (-3, 0)$

iii $(-5, 4)$ is a local maximum, and $(-3, 0)$ is local a minimum

d

i $f'(x) = 3x^2 + 30x + 75$

ii $(-5, 4)$

iii $(-5, 4)$ is a horizontal point of inflection

e

i $f'(x) = -x^2 + 13x - 30$

ii $\left(3, -\frac{61}{2}\right), \left(10, \frac{80}{3}\right)$

iii $\left(3, -\frac{61}{2}\right)$ is a local minimum, and $\left(10, \frac{80}{3}\right)$ is a local maximum

f

i $f'(x) = 8x + 9$

ii $\left(-\frac{9}{8}, -\frac{1}{16}\right)$

iii $\left(-\frac{9}{8}, -\frac{1}{16}\right)$ is a minimum turning point

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g

i $f'(x) = -300 + 480x - 192x^2$

ii $\left(\frac{5}{4}, 9\right)$

iii $\left(\frac{5}{4}, 9\right)$ is a horizontal point of inflection.

h

i $f'(x) = 4x^3 - 36x$

ii $(3, 4), (-3, 4), (0, 85)$

iii $(3, 4)$ and $(-3, 4)$ are minimum turning points and $(0, 85)$ is a local maximum

3

a $\left(\frac{1}{2}, \frac{21}{4}\right)$

b 0

c Minimum turning point

4

a $f'(x) = 2x + 4$

b $x > -2$

c $x < -2$

d $(-2, 5)$

e Maximum turning point

5

a $f'(x) = 12x^2 + 10x$

b $\left(-\frac{5}{6}, -\frac{307}{108}\right), (0, -4)$

c

x	-2	$-\frac{5}{6}$	$-\frac{1}{2}$	0	1
$f'(x)$	28	0	-2	0	22

d

i $(0, -4)$

ii $\left(-\frac{5}{6}, -\frac{307}{108}\right)$

e No, because the function is a cubic so it approaches $-\infty$.

6

a $f'(x) = 6x - 54$

b $x = 9$

c Minimum turning point

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